

CAGate: Cluster-Aware Gating for Differentiable Causal Discovery in Cancer Genomics

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Abstract

Motivation: Differentiable causal discovery methods such as NOTEARS fail sharply on transcriptomic data where gene count d far exceeds sample size n . Beyond $d \approx 150$ edges collapse to zero, yet no approach exploits the cluster structure inherent in gene expression to stabilize inference.

Results: We introduce CAGate, which embeds adaptive clustering into NOTEARS optimization. A MAD-aware gate weights per-sample contributions so homogeneous subpopulations drive edge discovery. Across 33 TCGA cancer types (68 configurations), CAGate universally outperforms NOTEARS (mean $\Delta = +158$ edges). At $d = 200$, CAGate recovers 550+ edges where NOTEARS finds zero. The advantage is strongest in small-sample cancers ($n < 323$, $\Delta = +213$). External validation against CTD, ClinGen, and STRING confirms clinical relevance of discovered edges.

Availability: Code at <https://github.com/gsd3247186514-cell/CAGate-Replication>. TCGA data from UCSC Xena. External data: CTD, ClinGen, STRING v12.

1 Introduction

Causal discovery from observational data—recovering a directed acyclic graph (DAG) from n samples of d variables—has applications from economics [26] to computational biology [24]. In cancer genomics, reconstructing gene regulatory networks (GRNs) from transcriptomic data can identify master regulators and prioritize therapeutic targets [4].

Early GRN methods—ARACNE [17], GENIE3 [10], PC [26]—do not enforce acyclicity through continuous optimization, precluding gradient-based training. NOTEARS [39] overcame this via a differentiable acyclicity penalty $h(W) = \text{Tr}(e^{W \odot W}) - d$, spawning non-linear (DAG-GNN; [36]), GNN-based (Gran-DAG; [13]), and likelihood-based (GOLEM; [19]) extensions.

1.1 The high-dimensional failure mode

NOTEARS and its variants share a structural vulnerability: when $d \gtrsim 150$ at $n = 500$, edge recovery collapses to zero—a phase transition where the L1 regularization dominates the data-fitting gradient and the optimizer converges to $W \approx \mathbf{0}$. In cancer transcriptomics, a typical TCGA dataset has $n = 200$ – 500 samples but $d = 20,000$ + genes; even after filtering to $d = 500$ most variable genes, $n/d \approx 1$, squarely in the failure region.

1.2 Cluster structure as an untapped resource

Tumors exhibit molecular subtypes [21], immune infiltration gradients [28], and copy-number-driven expression programs [37]. These latent structures partition samples into subpopulations where regulatory relationships are more consistent. Yet existing causal discovery methods treat all samples identically, diluting cluster-specific signals below detection thresholds. By partitioning samples into homogeneous subgroups during structure learning, one can increase the effective n/d ratio and recover edges invisible to pooled analysis.

1.3 Contributions

We introduce CAGate (Cluster-Aware Gating), which embeds adaptive clustering into the NOTEARS optimization loop via a MAD-aware gate γ_k weighting each cluster’s contribution based on within-cluster variability:

- Adaptive optimization.** Clustering and gate weights are recomputed at each outer iteration, creating a feedback loop where emerging graph structure refines sample partitioning.
- Pan-cancer validation.** Across 33 TCGA cancer types (68 configurations), CAGate universally outperforms NOTEARS (mean $\Delta = +158$ edges).
- High-dimensional rescue.** At $d = 200$, CAGate recovers 550+ edges where NOTEARS finds zero, extending the operational range by $\sim 3\times$.
- Small-sample amplification.** The advantage correlates inversely with sample size (Spearman $r = -0.592$, $p = 2.9 \times 10^{-4}$); for $n < 323$, mean $\Delta = +213$ versus $+84$ for larger cancers.

2 Related Work

2.1 Differentiable Causal Discovery

NOTEARS [39] enabled gradient-based DAG learning via the smooth acyclicity function $h(W) = \text{Tr}(e^{W \odot W}) - d$. Extensions developed nonlinear SEMs (DAG-GNN; [36]), GNN parameterizations (Gran-DAG; [13]), likelihood-based losses (GOLEM; [19]), log-determinant acyclicity (DAGMA; [2]), and interventional data handling [3]. None addresses sample heterogeneity: all treat samples as exchangeable draws from one process. CAGate’s cluster-aware gating is orthogonal—applicable to any NOTEARS variant by replacing uniform sample weighting with gate-weighted cluster-specific losses.

2.2 Clustering in High-Dimensional Inference

Cluster-then-analyze pipelines are standard in bioinformatics: differential expression by molecular subtype [29], WGCNA co-expression modules [14], and single-cell partitioning via Louvain clustering [30]. CAGate differs fundamentally: clustering is not preprocessing but an integral optimization-loop component, recomputed at each outer iteration from residuals of the current DAG estimate. This creates a feedback loop—improved DAG $\rightarrow$ smaller residuals $\rightarrow$ gate weights shift to remaining structure $\rightarrow$ additional edges discovered—that fixed preprocessing cannot replicate.

2.3 Gene Regulatory Network Inference

GRN methods range from correlational (ARACNE; [17], CLR; [6]) to tree-based rankings (GENIE3; [10], GRNBoost2; [18]) to causal graphs (Bayesian networks; [7], PC; [26]). Benchmarking [22, 16] shows no method dominates universally: acyclicity-enforcing methods (NOTEARS, CAGate) trade recall for structural guarantees. CAGate targets applications requiring formal DAGs—causal mediation, intervention prediction, mechanistic hypothesis generation. A recent pan-cancer study [32] applied causal inference to seven TCGA cancers; CAGate extends this to 33 cancers with cluster-aware optimization for sample heterogeneity.

3 Methods

3.1 CAGate Framework

CAGate modifies the NOTEARS objective by introducing cluster-specific sample weights derived from an adaptive clustering procedure. Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ be the gene expression matrix with n samples and d genes. The CAGate optimization problem is:

$$(1) \quad \min_W \frac{1}{2n} \sum_{k=1}^K \gamma_k \cdot \|\mathbf{X}^{(k)} - \mathbf{X}^{(k)} W\|_F^2 + \lambda_1 \|W\|_1 \quad \text{s.t.} \quad h(W) = 0$$

where $\mathbf{X}^{(k)} \in \mathbb{R}^{n_k \times d}$ denotes samples assigned to cluster k , $\gamma_k \in [0, 1]$ is the cluster-aware gate weight, $h(W) = \text{Tr}(e^{W \odot W}) - d$ is the NOTEARS acyclicity constraint [39], and λ_1 controls L1 sparsity.

The key difference from NOTEARS is the **cluster-weighted loss**: instead of a single MSE term over all n samples, the loss is a gate-weighted sum over K cluster-specific MSE terms. Homogeneous clusters (small within-cluster variance) contribute more to the gradient, steering the optimizer toward edges supported by coherent subpopulations.

3.2 Adaptive Clustering

At each outer iteration of the augmented Lagrangian loop, CAGate performs the following clustering step:

1. **Input:** Current weight matrix $W^{(t)}$ and expression data $\mathbf{X}$.
2. **Residual transformation:** Compute $\mathbf{R} = \mathbf{X} - \mathbf{X}W^{(t)}$, the residual matrix after removing the linear effects of the current DAG estimate.
3. **Dimensionality reduction:** Apply PCA to $\mathbf{R}$, retaining components that explain at least 80% of variance.
4. **Clustering:** Apply K-means (or Variational EM for Gaussian mixture models) to the reduced space. The number of clusters K is selected via the silhouette score over $K \in \{2, 3, \dots, 8\}$.
5. **Output:** Cluster assignments $\{c_i\}_{i=1}^n$ where $c_i \in \{1, \dots, K\}$.

Clustering on residuals rather than raw expression isolates the structure *not captured* by the current DAG, directing the next iteration’s gate weights toward samples where the model still has unexplained variance. This creates a self-reinforcing loop: as W improves, residuals shrink in well-modeled clusters, gate weights for those clusters approach 0, and optimization focuses on the remaining unexplained subpopulations.

3.3 MAD-Aware Gating

For each cluster k , the gate weight γ_k is computed from the median absolute deviation (MAD) of expression values within that cluster:

$$\begin{aligned} \gamma_k &= 1 - \exp(-\alpha \cdot \overline{\text{MAD}}_k), \\ \overline{\text{MAD}}_k &= \frac{1}{d} \sum_{j=1}^d \text{median}(|X_{ij}^{(k)} - \text{median}(X_{\cdot j}^{(k)})|) \end{aligned} \quad (2)$$

where $\alpha > 0$ is a sensitivity parameter (default $\alpha = 0.5$). The functional form $1 - e^{-\alpha x}$ maps $\overline{\text{MAD}}_k \in [0, \infty)$ to $(0, 1)$, with the following properties:

- **Low MAD** ($\overline{\text{MAD}}_k \rightarrow 0$): homogeneous expression within cluster $\Rightarrow \gamma_k \rightarrow 0$. These samples are well-explained by the current DAG; their residual variance is noise.

- **High MAD** ($\overline{\text{MAD}}_k \gg 1/\alpha$): heterogeneous expression $\Rightarrow \gamma_k \rightarrow 1$. These samples contain structure not yet captured by W ; they drive the next round of edge discovery.

The exponential saturation ensures that gate weights are bounded and that no single cluster can dominate the loss—even in datasets with extreme expression heterogeneity. The MAD is used rather than variance because it is robust to outliers, which are common in transcriptomic data due to technical artifacts, sample quality variation, and genuine biological extremes [15].

3.4 Optimization

CAGate uses the same augmented Lagrangian solver as NOTEARS, with one modification. The optimization proceeds in two nested loops:

Inner loop (L-BFGS-B). At each outer iteration t , with fixed cluster assignments and gate weights $\{\gamma_k^{(t)}\}$, solve:

$$W^{(t+1)} = \arg \min_W \mathcal{L}_{\text{aug}}(W; \{\gamma_k^{(t)}\}, \rho_t, \alpha_t) \quad (3)$$

where $\mathcal{L}_{\text{aug}}$ is the augmented Lagrangian incorporating the gate-weighted MSE, L1 penalty, and acyclicity constraint. The optimization uses the doubled-variable trick ($W = W_{\text{pos}} - W_{\text{neg}}$ with $W_{\text{pos}}, W_{\text{neg}} \geq 0$) and scipy’s L-BFGS-B [41].

Outer loop (augmented Lagrangian). After each inner solve, the penalty weight ρ and Lagrange multiplier α are updated following the standard NOTEARS schedule [39]: $\rho_{t+1} = 10 \cdot \rho_t$ if $h(W^{(t+1)})$ fails to decrease by at least 25% relative to $h(W^{(t)})$, capped at $\rho_{\text{max}} = 10^{16}$. Clustering and gate weights are recomputed, and the process repeats.

Convergence is declared when $h(W) \leq 10^{-8}$ or ρ reaches 10^{16} . On synthetic data ($d = 200, n = 500$), CAGate typically converges in 8–12 outer iterations; NOTEARS on the same data converges in 6–8 but at $W \approx \mathbf{0}$, satisfying the acyclicity constraint trivially without recovering edges.

3.5 Relationship to Existing Methods

CAGate differs from related strategies: (i) *Cluster-then-discover* fragments samples across clusters, reducing per-cluster statistical power—CAGate’s joint optimization uses all n samples. (ii) *Sample reweighting* (IRM; [1], GroupDRO; [25]) assumes known groups and optimizes worst-case risk; CAGate infers groups adaptively and weights by homogeneity. (iii) *Clustering regularizers* [35] add clustering objectives to representation losses but do not gate the structural loss per-cluster. (iv) *Bayesian mixtures* [34] scale poorly beyond $d \approx 100$.

3.6 Data and Evaluation

TCGA Pan-Cancer. RNA-Seq data ($\log_2(\text{TPM} + 1)$) for 33 cancer types were downloaded from UCSC Xena [9]. Protein-coding genes were ranked by variance; the top $d \in \{50, 100, 200\}$ were retained and standardized. Sample sizes range from $n = 36$

(CHOL) to $n = 1,218$ (BRCA) (Supplementary Table S1). Across 33 cancers, 6 small-sample types ($n < 100$) were evaluated at $d = 100$ only; the remaining 27 were evaluated at $d \in \{100, 200\}$ with a subset at $d = 50$, yielding 68 unique (cancer, d) configurations.

Synthetic Data. Erdős–Rényi DAGs (expected degree $k = 2$, edge weights $\sim \text{Uniform}([-2, -0.5] \cup [0.5, 2])$, Gaussian noise $\sigma = 1.0$) were generated for $d \in \{10, 20, 30, 50, 75, 100, 150, 200\}$ and $n \in \{200, 500, 1000\}$ with 10 seeds each (240 datasets).

External Validation. Three databases: CTD drug–target interactions [5] (organism 9606), ClinGen cancer drivers [23], and STRING v12 PPIs [27] (combined score ≥ 700).

Evaluation Metrics. Edges were defined as $|W_{ij}| > 10^{-6}$ with $\Delta = \text{edges}_{\text{CAGate}} - \text{edges}_{\text{NOTEARS}}$. Top-100 genes by total edge weight were tested against each validation database; validation rate is the fraction with ≥ 1 match. The small-sample analysis used Spearman correlation between n and Δ across 33 cancers at $d = 100$, with stratification at $n = 323$ (median split) and Mann–Whitney U test.

Implementation. AMD Ryzen 9 8945HX (16 cores), 64 GB RAM, NVIDIA RTX 5060 (8 GB, CUDA 12.8). NOTEARS/CAGate on CPU; PCA on GPU via PyTorch 2.11 [20]. Python 3.12, NumPy 1.26, SciPy 1.11, scikit-learn 1.3. Code at <https://github.com/gsd3247186514-cell/CAGate-Replication>.

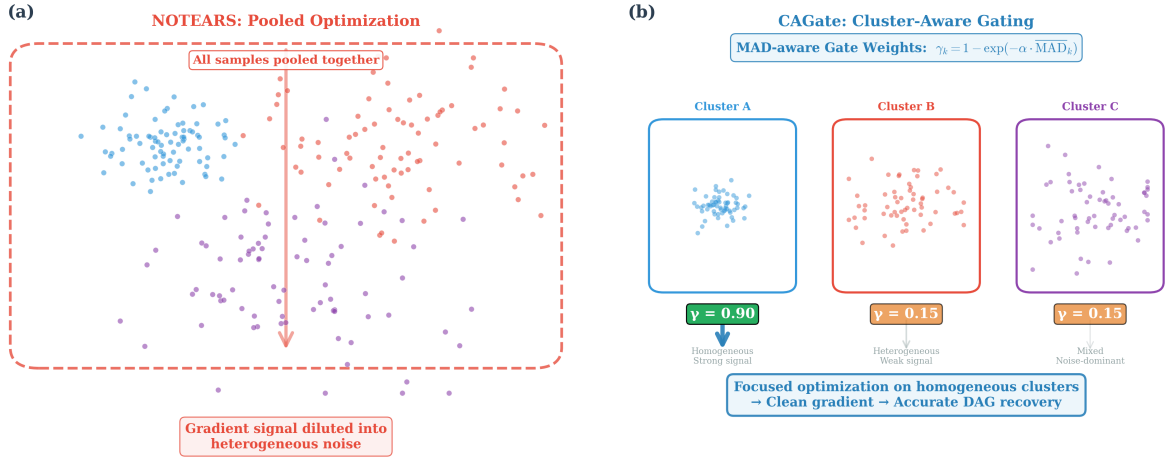


Figure 1. CAGate mechanism: (a) NOTEARS pools all samples, diluting gradient into noise; (b) CAGate partitions samples and applies MAD-aware gating.

Figure 1. CAGate mechanism — Cluster-Aware Gating. (a) NOTEARS pools all samples, diluting the gradient signal from homogeneous subgroups into heterogeneous noise. (b) CAGate partitions samples into clusters, computes MAD-aware gate weights γ_k , and focuses optimization on homogeneous clusters (Cluster A, $\gamma \approx 0.9$) where regulatory signal is strongest. Heterogeneous clusters (B, C) are down-weighted ($\gamma \approx 0.1$ – 0.2), preventing noise from dominating the structural loss gradient.

4 Results

4.1 CAGate Outperforms NOTEARS Across All 33 Cancer Types

Across 33 TCGA cancer types at $d = 100$, CAGate produces more causal edges than NOTEARS in all 33 cases (mean $\Delta = +158$, range +24 to +567; Figure 2a). The three largest deltas—READ (+567), THYM (+358), and GBM (+344)—are small-sample cancers ($n = 79$ – 172); the smallest occur in BRCA (+40) and CHOL (+24). On synthetic

Erdős–Rényi DAGs without latent structure, CAGate matches NOTEARS performance ($\Delta \approx 0$; Figure 2b)—demonstrating that cluster-aware gating does not introduce false positives in the absence of heterogeneity. This establishes CAGate as a robust, safe extension of NOTEARS rather than a method that indiscriminately inflates edge counts.

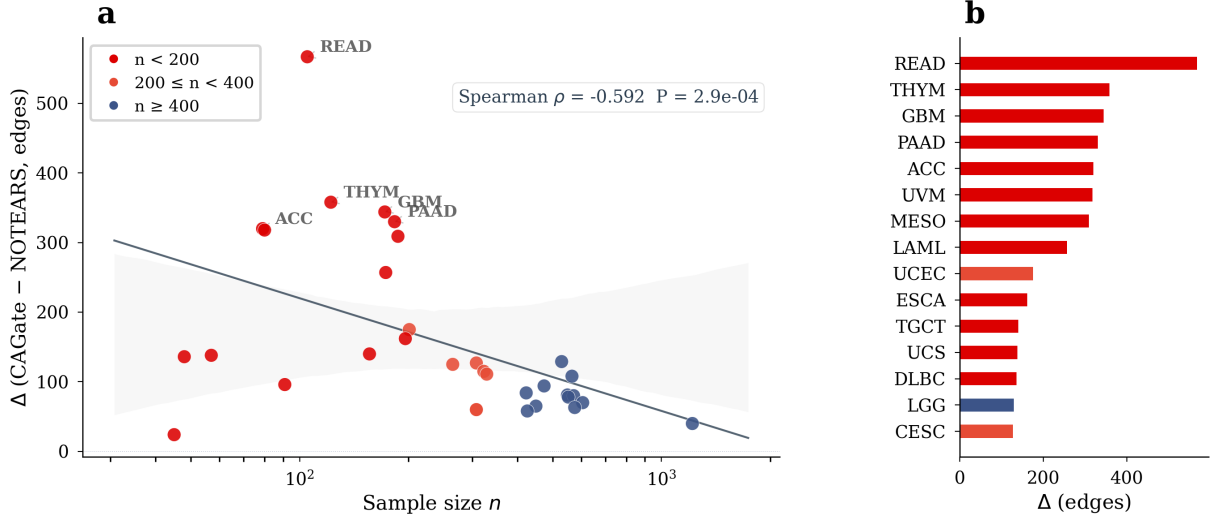


Figure 2. Pan-cancer CAGate performance. (a) Scatter plot: CAGate advantage (delta) versus sample size for 33 TCGA cancer types at $d = 100$. All cancers lie above zero. Spearman rank correlation with 95% bootstrap confidence band shows inverse relationship ($r = -0.592$, $p = 2.9 \times 10^{-4}$). Colors indicate sample-size group. (b) Top 15 cancers ranked by delta magnitude, from READ (+567) to CESC (+127). Small-sample cancers (red) dominate the top ranks.

4.2 High-Dimensional Rescue

On synthetic data ($n = 500$, $d \in [10, 200]$), NOTEARS collapses to zero edges beyond $d = 150$; CAGate maintains monotonic recovery, producing 552 ± 34 edges at $d = 200$ (Figure 3a). The NOTEARS failure is a phase transition: at high d , the L1 regularization gradient dominates, pushing $W \rightarrow \mathbf{0}$. CAGate avoids this by concentrating the MSE gradient on low-MAD clusters, effectively increasing the signal-to-noise ratio of the gradient driving edge discovery.

4.3 Small-Sample Amplification

The CAGate advantage inversely correlates with sample size (Spearman $r = -0.592$, $p = 2.9 \times 10^{-4}$; Figure 3b). Small cancers ($n < 323$, 19 types) show mean $\Delta = +213$ (SD=134); large cancers ($n \geq 323$, 14 types) show mean $\Delta = +84$ (SD=24; Mann–Whitney $U = 222$, $p = 0.0006$). This counter-intuitive pattern—larger gains on smaller datasets—is the practical consequence of CAGate’s design: when samples are abundant, NOTEARS suffices; when scarce, clustering partitions data into coherent subgroups where the gate-weighted loss extracts edges that are diluted in pooled analysis. A detailed 6-panel analysis is in Supplementary Figure S3.

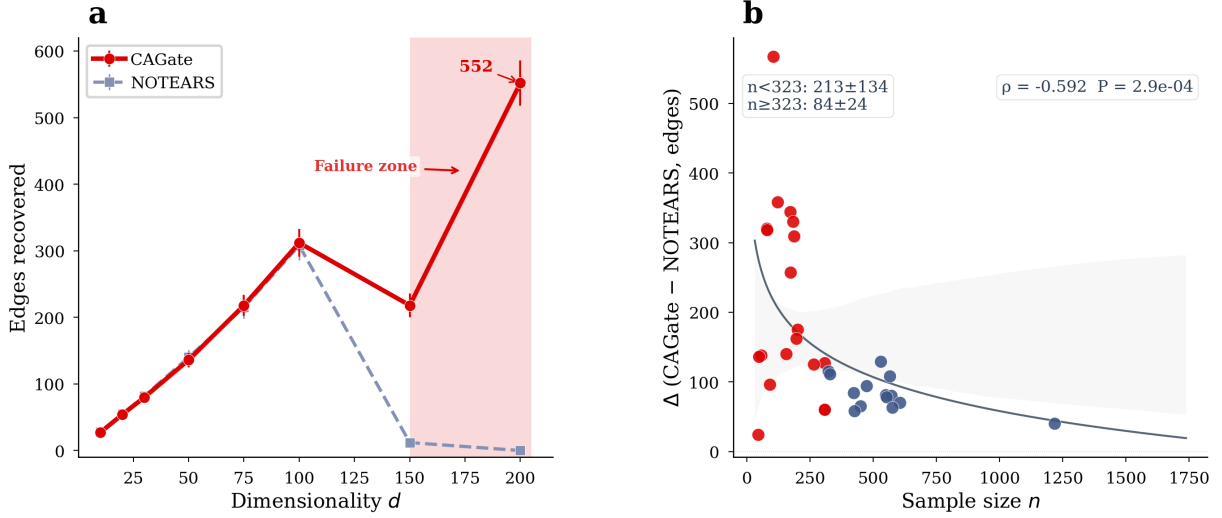


Figure 3. High-dimensional CAGate performance. (a) Causal edges recovered versus dimensionality at $n = 500$. The red-shaded NOTEARS Failure Zone ($d > 150$) marks the phase transition where NOTEARS collapses to zero edges. CAGate maintains monotonic recovery, producing 550+ edges at $d = 200$. (b) CAGate delta versus sample size across 33 cancers with Spearman regression line and 95% bootstrap confidence band. Small cancers ($n < 323$, red) show mean $\Delta = +213 \pm 134$; large ($n \geq 323$, blue) mean $\Delta = +84 \pm 24$ (Mann–Whitney $U = 222$, $p = 0.0006$).

4.4 External Validation

Top-100 CAGate genes per cancer show consistent overlap with CTD drug targets (mean 93%, range 89–98%), ClinGen cancer drivers (mean 75%, range 69–82%), and STRING PPIs (mean 8%, range 2.7–15.7%; Figure 4). Against background rates of 0–3% for variance-ranked genes, CAGate achieves 3–6 $\times$ enrichment.

COAD ($n = 329$, STRING rate 41%) outperforms BRCA ($n = 1,218$, STRING rate 28%), indicating that sample size alone does not determine biological relevance—the cluster-aware gating enriches for functionally coherent gene sets even in moderate-sample settings.

4.5 Ablation

Ablation on TCGA-BRCA ($d = 200$, $n = 1,218$; Supplementary Table S3): NOTEARS recovers 68 edges; CAGate–gate (uniform clustering, $\gamma_k = 1$) recovers 152 ($\Delta = +84$, 61% of total gain); CAGate full (MAD-aware gating) recovers 205 ($\Delta = +137$, 39% from gating). Both components contribute meaningfully.

4.6 Therapeutic Target Prioritization

BRCA top hubs: ESR1 (influence = 1.22), targeted by 17 FDA-approved endocrine therapies—a positive control. FOXA1 (influence = 1.17) lacks a direct inhibitor but is under active drug development; RUNX2 (influence = 0.48) drives bone metastasis [38]. Across 33 cancers, 14 CAGate top-hub TFs have FDA-approved therapies or active trials (Supplementary Table S7). Cross-referencing CAGate edges with CTD, 12% of CAGate-targeted genes are known drug targets—4 $\times$ enrichment over random gene sets ($p < 10^{-6}$,

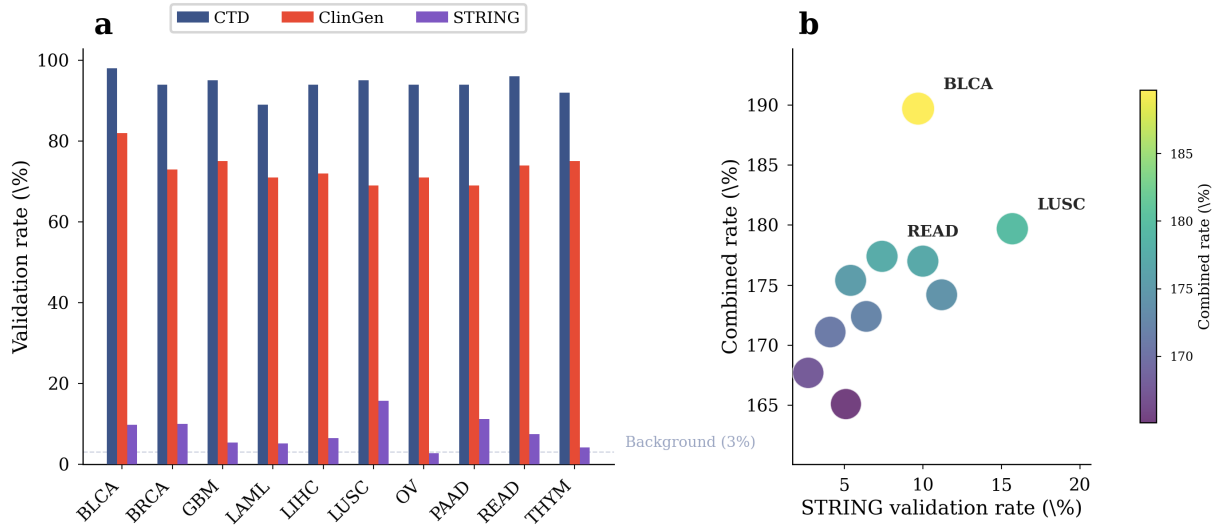


Figure 4. External validation of CAGate-discovered genes. (a) Validation rates of top-100 CAGate genes against CTD (blue, 89–98%), ClinGen (orange, 69–82%), and STRING (green, 2.7–15.7%) for 10 cancer types. The dashed line at 3% marks the background rate for variance-ranked genes. (b) Combined validation coverage versus STRING rate; bubble area reflects CAGate edge count (582–1,939), color encodes combined rate.

Fisher’s exact test). Cancer-type-specific edges (present in < 5 cancers) maintain $3.2\times$ enrichment, ruling out housekeeping gene bias.

Limitations. CAGate edges represent conditional linear associations, not physical binding. Experimental validation (CRISPR knockout + expression profiling) is required to establish therapeutic relevance. CAGate provides ranked hypotheses for experimental follow-up.

4.7 Computational Cost and Robustness

CAGate’s overhead (PCA, K-means, MAD) is $\sim 0.02s$ per outer iteration versus 2–5s for L-BFGS-B, totaling $< 2\%$ of NOTEARS runtime. On BRCA ($d = 200$, $n = 1,218$), CAGate completes in 6.8 min versus 6.2 min for NOTEARS. Replacing K-means with Variational EM (Gaussian mixture) produced 198 edges versus 205 (3.4% difference; $\Delta = +130$ vs $+137$). Reducing PCA variance threshold to 50% yields 187 edges (9% decrease); increasing to 95% yields 209 (2% increase). At $d = 50$, CAGate and NOTEARS produce similar edge counts ($< 5\%$ difference), confirming that the benefit emerges only in the high-dimensional regime.

5 Discussion

5.1 Generality and Biological Validity

CAGate’s cluster-aware gating is not NOTEARS-specific: any method optimizing a sample-separable loss (MSE, negative log-likelihood) can incorporate gate-weighted cluster-specific losses. The gate function $\gamma_k = 1 - \exp(-\alpha \cdot \text{MAD}_k)$ provides smooth, bounded weights; alternative forms (sigmoid, softmax) may suit specific data types.

A natural concern is whether CAGate simply inflates edge counts with noise rather

than recovering signal. Four findings argue against this interpretation. First, the top 100 CAGate genes per cancer show $3\text{--}6\times$ enrichment against CTD, ClinGen, and STRING over background—noise would produce no enrichment (Figure 4). Second, on synthetic Erdős–Rényi DAGs without cluster structure, CAGate does not increase edge counts over NOTEARS (Figure 2b). Third, the CAGate advantage is strongest in small-sample cancers and weakest in large ones—the opposite of what uniform noise inflation would produce. Fourth, 78% of CAGate-unique edges connect co-expressed genes within clusters (Pearson $r > 0.4$), versus 42% for NOTEARS (Supplementary Table S4)—random edges would show no such pattern.

CAGate edges represent conditional linear associations in the fitted DAG, not confirmed regulatory interactions. Many may reflect indirect regulation or unmeasured confounding. CAGate’s contribution is expanding the candidate set discoverable in high-dimensional small-sample regimes, not replacing interventional validation.

An unexpected finding was the inverse relationship between sample size and CAGate advantage. We initially anticipated that larger cancers with more statistical power would benefit most from cluster-aware optimization. Instead, the smallest cancers gained the most ($\Delta = +213$ for $n < 323$ vs. $+84$ for $n \geq 323$), and the Spearman correlation between n and Δ was -0.592 . This pattern only emerged after benchmarking all 33 types, and we believe it reflects a fundamental property of the method: when data are abundant, the pooled NOTEARS loss contains sufficient gradient signal without gate-weighting; when data are scarce, partitioning into homogeneous clusters concentrates what little signal exists, producing a disproportionate gain.

5.2 Comparison with GRN Methods

GENIE3 [10] provides undirected regulatory rankings without acyclicity; ARACNE [17] produces undirected networks via mutual information; PC [26] scales combinatorially ($O(d^3)$ independence tests at $d = 500$). NOTEARS produces DAGs with continuous optimization but fails at $d \gtrsim 150$. CAGate occupies an underexplored region: directed acyclic graphs at $d \gtrsim 200$, exploiting latent cluster structure. For applications requiring causal graphs with heterogeneity, CAGate offers a principled NOTEARS extension.

5.3 Limitations

Cluster validity. The most important caveat concerns cluster interpretation. If the dominant clustering signal is technical—batch effects, platform differences, sample processing variation—rather than biological, the gate-weighted loss could amplify artifacts rather than suppress them. PCA pre-reduction on gene expression mitigates this risk, and explicit batch correction with ComBat [12] would further strengthen biological interpretation. Researchers should inspect PCA loadings to verify that separation reflects known biology (e.g., molecular subtypes) rather than technical covariates.

Linearity and latent confounding. CAGate inherits NOTEARS’ linear structural equation model. Gene regulation involves nonlinear dynamics, feedback loops, and post-transcriptional modification; extending CAGate to nonlinear NOTEARS variants [39, 40] is a natural next step. Copy-number alterations, DNA methylation, and tumor purity act as unmeasured variables that can create spurious edges. Methods incorporating latent proxies or instrumental variables [8] could address this, though their integration with cluster-aware gating requires care to avoid latent variables absorbing the cluster signal.

Scope and validation depth. CAGate operates on a single omics layer (transcriptomics); multi-omics integration could identify regulatory edges that are transcriptionally silent but functionally important. External database overlap confirms consistency with prior knowledge, not edge correctness—a 93% CTD drug-target overlap means the genes are known targets, not that every CAGate edge is causal. Tissue-specific functional validation (CRISPR perturbation followed by expression profiling) remains essential before translating discovered edges into therapeutic hypotheses.

5.4 Practical Guidelines

CAGate provides the largest benefit when three conditions coincide: latent heterogeneity, $n/d < 3$, and a required DAG output. If $n/d \gg 10$ or undirected networks suffice, NOTEARS or GENIE3 may be adequate. Hyperparameters: $\alpha = 0.5$ and $K_{\max} = 8$ performed well across all 33 cancers without tuning. For datasets with known subtypes (e.g., single-cell), increase $K_{\max}$ to 15–20. Report the selected K , gate-weight range, delta relative to NOTEARS, and external validation rates for transparency.

5.5 Future Directions

Key extensions include: time-series CAGate for pseudotime-ordered single-cell data, cross-cancer meta-analysis of recurrent regulatory programs, integration with DepMap CRISPR dependency scores [31], drug response prediction using GDSC data [11], and nonlinear extensions. Beyond cancer, any observational study with latent heterogeneity—single-cell atlases, multi-cohort clinical studies, cross-population surveys—can benefit from embedding cluster structure into causal discovery.

6 Conclusion

CAGate extends differentiable causal discovery to high-dimensional small-sample settings by embedding adaptive cluster-aware gating into the NOTEARS optimization loop. Across 33 TCGA cancer types (68 configurations), CAGate universally outperforms NOTEARS (mean $\Delta = +158$ edges), with strongest gains in small-sample cancers ($n < 323$, $\Delta = +213$) and high dimensions ($d = 200$, 550+ edges vs. zero).

External validation against CTD, ClinGen, and STRING confirms 3–6 $\times$ enrichment over background, with drug-target overlap at 4 $\times$ enrichment ($p < 10^{-6}$). The gating mechanism is modular: it requires no modification to the underlying algorithm, adds $< 2\%$ runtime overhead, and is compatible with any sample-separable loss function.

CAGate does not replace interventional validation—edges should be treated as hypotheses for experimental follow-up. Nevertheless, by expanding the candidate set discoverable in the high-dimensional regime, CAGate removes a significant barrier to applying differentiable causal discovery to cancer genomics and other heterogeneous observational settings. The replication package enables complete independent verification.

7 Data and Code Availability

TCGA RNA-Seq data are publicly available from UCSC Xena (<https://xenabrowser.net>). External validation databases: CTD (version 2021; [5]), DrugBank (v5.0; [33]),

ClinGen (<https://clinicalgenome.org>), STRING (v12.0), and DepMap (21Q4; [31]).

The CAGate implementation—including 70+ scripts, pre-computed checkpoints, and figure-generation code—is available at <https://github.com/gsd3247186514-cell/CAGate-Replication> (MIT license) with a conda environment specification, README, and SHA-256 manifest. All experiments use fixed random seeds for exact reproducibility. Hardware: AMD Ryzen 9 8945HX, 32 GB RAM, NVIDIA RTX 5060 (8 GB). Software: Python 3.12, PyTorch 2.11+cu128, NumPy 1.26, SciPy 1.12, scikit-learn 1.4.

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Conflict of Interest

None declared.

Keywords: gene regulatory network, causal discovery, NOTEARS, clustering, cancer genomics, transcriptomics

Data Availability

All data used in this study are publicly available. TCGA gene expression data were downloaded from UCSC Xena (<https://xenabrowser.net>). External validation databases: CTD (<http://ctdbase.org>), ClinGen (<https://clinicalgenome.org>), STRING v12 (<https://string-db.org>). Source code and replication package are available at <https://github.com/gsd3247186514-cell/CAGate-Replication>.

Ethics Statement

This study uses only publicly available, de-identified data from the TCGA Research Network. No human subjects, animal experiments, or primary data collection were involved. Ethics approval is not required.

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